

COMMON ANKLE & FOOT DEFORMITIES

Terminology

- Normal foot = **plantigrade** (sole at right angle to leg).
- **Equinus** = hindfoot fixed plantarflexion (horse foot). **Plantaris** = forefoot plantarflexed, ankle neutral. **Calcaneus** = fixed dorsiflexion. **Rocker-bottom** = mid-foot dorsiflexion deformity.
- **Planus** (flat-foot) = flattened longitudinal arch; **cavus** = excessively high arch.
- **Hallux valgus** = lateral deviation of big toe. **Hammer-toe** = PIP flexion. **Claw toe** = IP flexion + MTP hyperextension.

Congenital Talipes Equinovarus (Clubfoot / TEV)

- Commonest = **idiopathic TEV**. Deformity: **heel equinus + hindfoot varus + mid/forefoot adducted + supinated**.
- **Incidence 1–2/1000; boys 2:1; bilateral 1/3**. Calf + medial soft tissues short/underdeveloped.
- **Causes**: idiopathic (commonest); neurological (polio, cerebral palsy); **spina bifida (myelomeningocele)**; trauma/burn; arthrogryposis.
- **Clinical**: exclude in every neonate. **True clubfoot = big toe CANNOT reach shin; postural = big toe CAN reach shin (resolves)**. **Examine spine for spina bifida**.
- **X-ray**: tarsals incompletely ossified; **talus + calcaneum first to ossify** (3 months before birth); assesses progress.
- **Treatment: start within 1–2 days of birth. Ponseti method** (international standard):
- Correct all components simultaneously **EXCEPT equinus (corrected LAST)** — otherwise rocker-bottom deformity (vertical talus → rigid flat foot).
- Cavus corrected by abducting supinated forefoot; heel varus auto-corrects when fully abducted under talus (counter-pressure on talar head).
- **Equinus last** → dorsiflex ± **percutaneous Achilles tenotomy**.
- **Weekly POP casts ×5** (groin to MTP) → Denis Browne brace / ankle-foot orthosis. **Relapse common** (esp. neuromuscular).
- Resistant/relapsed → surgery / **Ilizarov** (circular external fixator).

Hallux Valgus

- **Outward (lateral) deviation of big toe** — commonest foot deformity. **Metatarsus primus varus** (congenital or acquired elderly — ↓muscle tone).
- **Clinical**: pain over **bunion** (prominence over 1st metatarsal head, medial); great toe in valgus.
- **X-ray (standing)**: intermetatarsal angle (1st–2nd MT) normally **<9°**; hallux valgus angle (1st MT–phalanx) **<15°**.
- **Treatment: flat shoes** (better than raised heel); surgery if severely painful. **Don't operate for cosmesis** (high recurrence + pain transfer).

High-yield one-liners

- **Clubfoot = equinus + varus + adduction + supination; idiopathic commonest; boys 2:1; bilateral 1/3**.
- **True vs postural clubfoot: big toe reaches shin = postural (resolves)**. Examine spine (spina bifida).

- **Ponseti method: correct all except equinus, equinus LAST (avoid rocker-bottom); weekly casts + Achilles tenotomy; relapse common.**
- **Hallux valgus = lateral big toe deviation + bunion; intermetatarsal angle $<9^\circ$, HV angle $<15^\circ$ normal; flat shoes, surgery only if painful (not cosmetic).**

COMMON ANKLE & FOOT DEFORMITIES (Revision Table)

Bold = high-yield. Prof. Omer Barawi.

TERMINOLOGY	
Term	Meaning
Plantigrade	Normal (sole ⊥ leg)
Equinus / Calcaneus	Hindfoot plantarflexion / fixed dorsiflexion
Rocker-bottom	Mid-foot dorsiflexion (vertical talus)
Planus / Cavus	Flat / high arch
Hallux valgus / hammer / claw	Lateral big toe / PIP flexion / IP flexion + MTP hyperextension

CLUBFOOT (Talipes Equinovarus)	
Aspect	Key points
Deformity	Heel equinus + hindfoot varus + forefoot adduction + supination
Epidemiology	1–2/1000; boys 2:1; bilateral 1/3
Causes	Idiopathic (commonest); spina bifida, CP, polio, arthrogryposis
Clinical	True = big toe CANNOT reach shin; postural = CAN reach (resolves); examine spine
Ponseti method	Start 1–2 days of birth; correct all EXCEPT equinus → equinus LAST (avoid rocker-bottom); weekly casts ×5 + Achilles tenotomy → brace; relapse common
Resistant/relapsed	Surgery / Ilizarov

HALLUX VALGUS	
Aspect	Key points
Definition	Lateral big toe deviation; commonest foot deformity; metatarsus primus varus
Clinical	Pain over bunion (1st MT head, medial); great toe valgus
X-ray (standing)	Intermetatarsal angle <9°, HV angle <15° normal
Treatment	Flat shoes; surgery if painful; not for cosmesis (recurrence + pain transfer)